

AO-SOC Command Center — User Manual

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This manual is for the **people who use AO-SOC**: the SOC analyst on shift, the shift lead, and whoever installs it. It covers how to install it, how to sign in, what each menu and page shows, and how to work an incident from the queue to the archive.

It is a how-to guide. For design, APIs and configuration details, see [README.md](#) , [orchestrator/README.md](#) and the staged rollout in [docs/PILOT-RUNBOOK.md](#) .

All screenshots come from the built-in demo (Section 2.1), so the incidents shown are sample data.

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1. What AO-SOC is, in one minute

AO-SOC is the **decision layer** of a Security Operations Center. It does **not** collect logs, store logs or detect attacks. Your existing tools do that (Splunk, Wazuh, Elastic, Sentinel, CrowdStrike, or anything that sends CEF). AO-SOC does four things with what those tools send it:

1. **Correlates.** Detections from different tools about the same user, host or address in the same time window become one **Security Situation**. For example, a Splunk brute-force alert, a Wazuh privilege escalation and a firewall egress hit on the same account become one incident, not three.
2. **Decides.** A local AI model (run through Ollama, fully on-prem) reads the situation and proposes a **Tier-2 decision**: `CONTAIN` , `ESCALATE` , `INVESTIGATE` , `MONITOR` or `IGNORE` . It also proposes a bundled **action plan**, such as *Block IP*, *Isolate host* or *Disable account*.
3. **Asks a person.** You review the plan once and click **Approve plan**, **Edit** or **Reject**.
4. **Executes and records.** Approved actions go to your response tools (SOAR, EDR, firewall, identity provider). Every step is audited, and an action that can be undone can be **rolled back** from the dashboard.

Detection tools → AO-SOC correlation → AI Tier-2 decision → YOU (approve / edit / reject)
→ policy guardrails → response tools → audit, archive, rollback

Playbooks in AO-SOC. In this product a *playbook* is the **action plan attached to each decision**. It is visible in two places: the **Bundled action plan** in the AI Tier-2 Decision panel, where you approve it and it runs, and the **Incident Playbook** checklist on the Live Alerts page, which is a read-only containment summary. You do not write playbooks by hand. The AI proposes one for each incident, and you approve it, correct it or reject it.

2. Installation

There are two ways to run AO-SOC:

	Use it for	Section
Demo	Trying it out, training, recordings. Sample alerts, no real model needed	2.1
Production (Docker)	A real pilot or deployment with real detections	2.2

2.0 Requirements

Component	Version	Needed for
Python	3.11 or later	Broker (demo)
Node.js + npm	20 or later	UI API and dashboard (demo)
Docker + Docker Compose	recent	Production
Ollama + model <code>qwen3.5:latest</code> (or <code>qwen2.5:7b</code>)	optional	Real AI decisions. Without a model, AO-SOC runs in <i>model-free</i> mode and a rule fallback decides
Browser	Current Edge, Chrome or Firefox	Dashboard. Designed for 1920×1080 and larger; works on tablets and phones

Ports

Service	Port	Notes
Broker (AI decision engine)	8500	Receives detections
UI API	4317	Serves the dashboard's data
Dashboard (demo, Vite)	5173	<code>http://localhost:5173</code>
Dashboard (production, Docker)	8080	The only port Docker publishes. Change it with <code>DASHBOARD_PORT</code>

2.1 Demo installation (Windows, Linux, macOS)

Step 1. Get the code and open a terminal in the `ao-soc` folder.

Step 2. Start everything with one command.

Windows (PowerShell):

```
.\scripts\start-demo.ps1
```

Linux / macOS:

```
chmod +x scripts/start-demo.sh scripts/stop-demo.sh
./scripts/start-demo.sh
```

The script:

1. checks that Python and Node are installed,
2. installs the dependencies (`-SkipInstall` / `--skip-install` skips this on later runs),
3. starts the broker, the UI API and the dashboard, each in its own window,
4. loads **12 sample alerts**, including one situation correlated across three tools,
5. **prints the operator key you sign in with.**

Sign in at the dashboard with this operator key:
Xy7...your-key...Qp

Copy the key. You need it in [Section 3](#).

Step 3. Open <http://localhost:5173> in your browser.

Demo variations

Command (Windows / Linux)	What you get
<code>start-demo.ps1</code> / <code>start-demo.sh</code>	12 alerts loaded at once
<code>start-demo.ps1 -Live</code> / <code>start-demo.sh --live</code>	Alerts arrive live over about 2 minutes, so you can watch the queue fill
<code>start-demo.ps1 -Ai -Count 6</code> / <code>start-demo.sh --ai --count 6</code>	Real AI: the local Ollama model analyses each alert. Budget 10–40 s per alert

Stop the demo:

```
.\scripts\stop-demo.ps1
```

```
./scripts/stop-demo.sh
```

Each demo start **resets** the demo alerts. Do not run the demo scripts against a production database.

2.2 Production installation (Docker)

Production follows the staged rollout in [docs/PILOT-RUNBOOK.md](#). In short:

Step 1. Create the settings file.

```
cp deploy/.env.example deploy/.env
```

Step 2. Generate one key per person and per system. Run this once for each key:

```
python -c "import secrets; print(secrets.token_urlsafe(32))"
```

Fill them into `deploy/.env` :

Setting	Who holds the key	Example
<code>BROKER_API_KEYS</code>	Machines: the SIEM (<code>ingest</code>), the UI API (<code>service</code>)	<code>ui-api:service:<key>,siem:ingest:<key></code>
<code>UI_API_BROKER_KEY</code>	The UI API: the same secret as its <code>service</code> entry above	<code><key></code>
<code>AOSOC_API_KEYS</code>	Each analyst personally , for signing in to the dashboard	<code>soclead:analyst:<key>,dutydesk:viewer:<key></code>

The format is `name:role:secret` , with entries separated by commas. **Give every analyst their own key.** Approvals, edits and rollbacks are recorded under the key's name, so a shared key leaves an audit trail that names nobody.

Step 3. Check the settings, then start.

```
python deploy/check_env.py
```

```
docker compose --env-file deploy/.env -f deploy/docker-compose.yml up -d
```

`check_env.py` warns when a variable left in your shell (for example from a demo) silently overrides `deploy/.env` .

Step 4. Open the dashboard at `http://<server>:8080` and sign in with your analyst key.

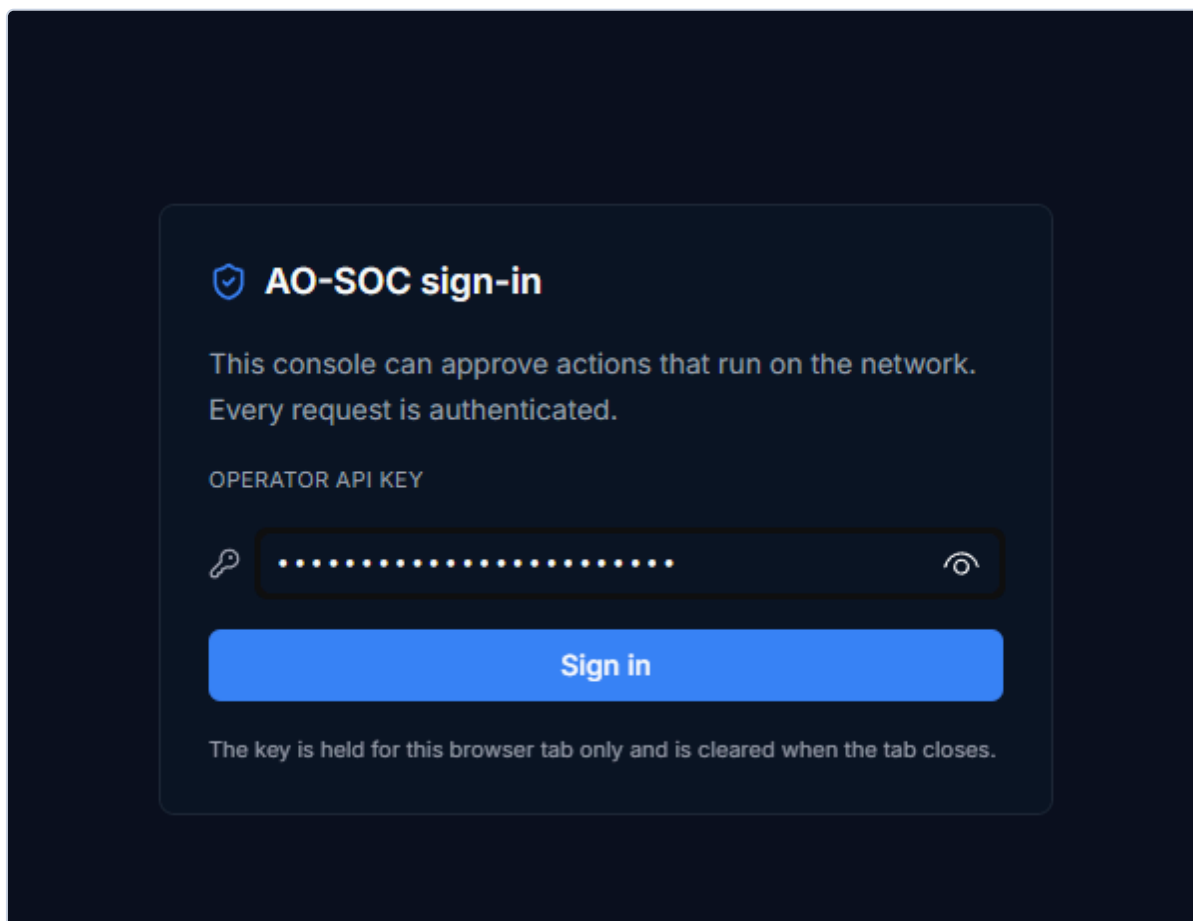
Step 5. Point your detection tools at the broker. For example, Splunk sends to `POST http://<broker-host>:8500/splunk-alert` , and any other vendor sends to `POST http://<broker-host>:8500/detections` . Each request carries the **ingest** key in the `X-API-Key` header. A detection sent without a key is rejected.

Safe defaults. A fresh install starts with `LLM_PROVIDER=echo` (no model), `TIER2_AUTOPILOT=false` (a person approves everything) and `RESPONSE_DRY_RUN=true` (nothing is sent to any real tool). Turn these on one at a time, as the runbook describes.

2.3 Roles

Role	Can do
<code>viewer</code>	See everything. Cannot approve, edit, reject or roll back
<code>analyst</code>	See and act: approve, edit, reject, record outcomes, roll back, work cases
<code>ingest</code>	Machine role: may only send detections
<code>service</code>	Machine role: the UI API acting for the signed-in analyst
<code>admin</code>	Everything. Do not hand it out by default

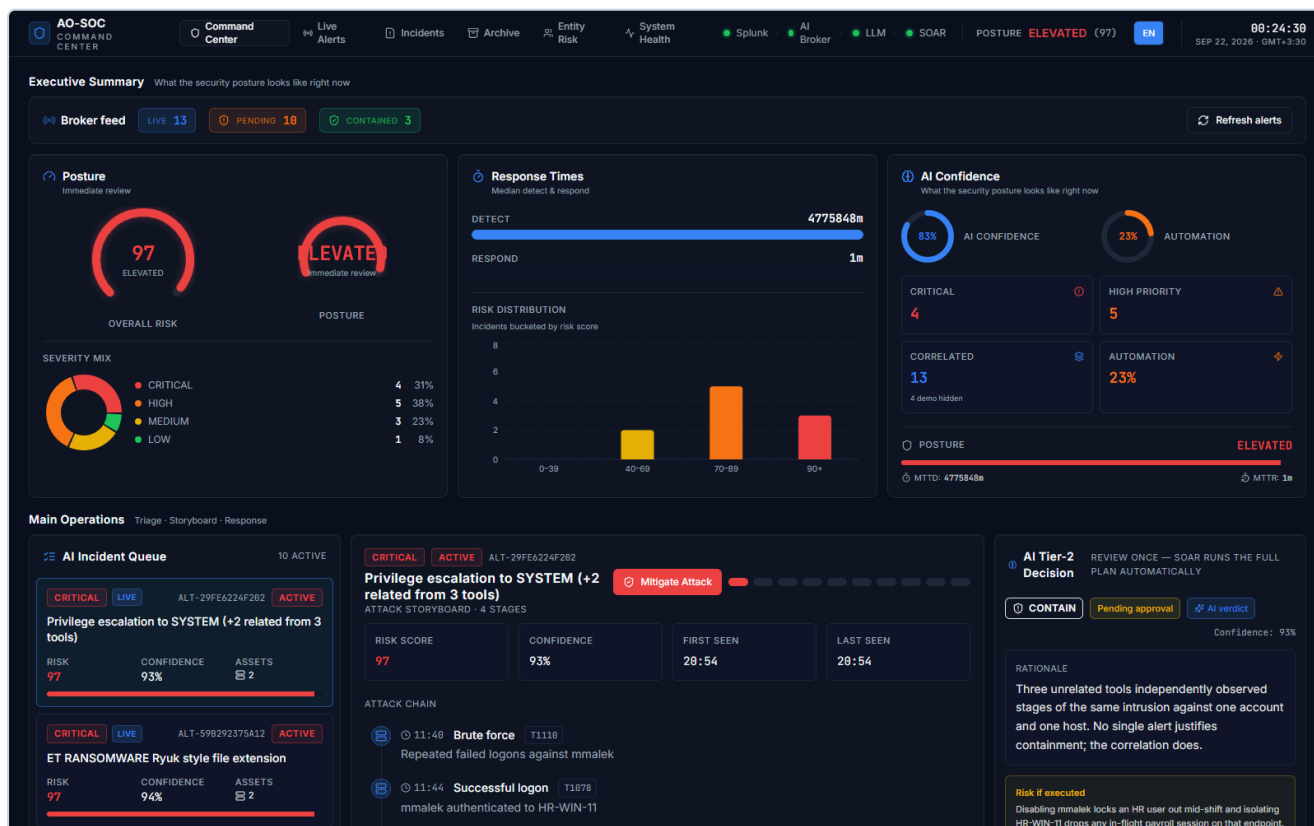
3. Signing in



1. Open the dashboard URL.
2. Paste your **Operator API key** into the box.
3. Click **Sign in**.

The key is kept **for this browser tab only**. Closing the tab signs you out, so a shared SOC workstation does not stay signed in for the next shift. If the key is wrong or has been revoked, the sign-in screen comes back.

4. The screen layout and the top bar



Every page has the same **top bar**:



#	Element	What it does
1	AO-SOC logo	Returns to the Command Center
2	Main menu	The six pages (table below)
3	Pipeline status (Splunk · AI Broker · LLM · SOAR)	A green dot means the part is online, amber degraded, red offline, grey unknown. Refreshed every 5 seconds
4	Posture	Overall security posture (LOW / GUARDED / HEIGHTENED / ELEVATED) with its score from 0 to 100
5	EN / FA	Switches the language between English and Persian (right-to-left layout)
–	Clock	Local time and date with the time-zone offset

The main menu

Menu item	Page	Use it to...
Command Center	/	See everything at once and work incidents. This is your home screen
Live Alerts	/alerts	Scan the raw alert log and see each alert's containment playbook
Incidents	/incidents	See the full list of active incidents and open one in detail

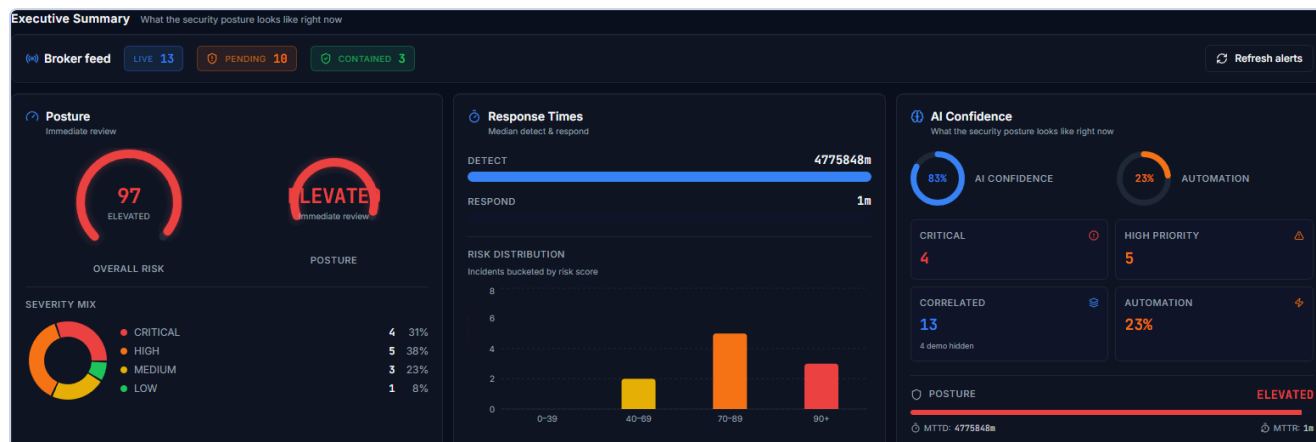
Menu item	Page	Use it to...
Archive	/archive	Review cleared incidents: what was decided, who approved it, what ran
Entity Risk	/entities	See the riskiest users, hosts and IP addresses
System Health	/health	Check that the pipeline is alive and look at its telemetry

The **footer** shows the application version (for example `v2.8.8`) and the pipeline `Splunk → AI Broker → Local LLM (Qwen) → SOAR`. Quote the version when you report a problem.

5. Command Center (home page)

The Command Center answers five questions from top to bottom: *What needs attention now? How serious is it? Why does the AI think so? What should be done? What is our posture?* It has six rows.

5.1 Executive Summary: "how are we doing right now?"



- **Broker feed.** Counts of **LIVE** incidents, incidents **PENDING** a human decision, and **CONTAINED** ones. **Refresh alerts** reloads them immediately. The page also refreshes by itself every 15 seconds.
- **Posture.** The overall risk gauge (0–100) with its label, plus the **Severity Mix** donut of open incidents.
- **Response Times.** **MTTD** (mean time to detect) and **MTTR** (mean time to respond), plus the **Risk Distribution** histogram of incidents by risk score.
- **AI Confidence.** The average model confidence, the automation rate, and counts of **Critical**, **High Priority** and **Correlated** incidents.

AI confidence is shown, never used to decide. Local models report 75–98 % confidence almost regardless of the input. Use the number to sort your work, not as a reason to approve.

5.2 Main Operations: triage, storyboard and response

This is where you work incidents.

AI Incident Queue

AI Incident Queue

10 ACTIVE

CRITICAL

LIVE

ALT-29FE6224F2B2

ACTIVE

Privilege escalation to SYSTEM (+2 related from 3 tools)

RISK

97

CONFIDENCE

93%

ASSETS

2

CRITICAL

LIVE

ALT-59B292375A12

ACTIVE

ET RANSOMWARE Ryuk style file extension

RISK

97

CONFIDENCE

94%

ASSETS

2

CRITICAL

LIVE

ALT-2AF2A4D8B8DF

ACTIVE

ET EXPLOIT Possible EternalBlue SMB attempt

RISK

97

CONFIDENCE

94%

ASSETS

2

HIGH

LIVE

ALT-7C421F761BCD

ACTIVE

ET INFO Suspicious TLS SNI to DGA-like domain

RISK

78

CONFIDENCE

88%

ASSETS

2

HIGH

LIVE

ALT-F8C48E86CB0

ACTIVE

ET DNS Query for .onion TLD

RISK

78

CONFIDENCE

88%

ASSETS

2

HIGH

LIVE

ALT-2DC746149A2F

ACTIVE

ET POLICY Suspicious inbound to MSSQL port 1433

RISK

78

CONFIDENCE

88%

ASSETS

2

HIGH

LIVE

ALT-731103170BF4

ACTIVE

ET POLICY Powershell DownloadString

RISK

78

CONFIDENCE

88%

ASSETS

2

HIGH

LIVE

ALT-011AEDAA66EC

ACTIVE

ET MALWARE Known C2 Beacon

RISK

78

CONFIDENCE

88%

ASSETS

2

MEDIUM

LIVE

ALT-FEC2C8E326B6

ACTIVE

ET WEB_SERVER SQL Injection Attempt

RISK

62

CONFIDENCE

71%

ASSETS

2

MEDIUM

LIVE

ALT-27B5C54BD465

ACTIVE

ET SCAN Potential SSH Scan

RISK

62

CONFIDENCE

71%

ASSETS

2

Each card shows the **severity** (CRITICAL , HIGH , MEDIUM , LOW), a **LIVE** badge (real broker data; DEMO marks sample data), the incident id, its **status**, the title, the **risk** score, the AI **confidence** and the number of affected **assets**. The bar at the bottom of each card is the risk score, coloured by severity.

A title such as "*Privilege escalation to SYSTEM (+2 related from 3 tools)*" means AO-SOC joined three detections from three different tools into one situation.

Attack storyboard

The interface displays an incident titled "Privilege escalation to SYSTEM (+2 related from 3 tools)" with a risk score of 97, 93% confidence, and a first/last seen time of 20:54. It includes a "Mitigate Attack" button and a progress bar. The attack chain shows four stages: Brute force (T1110), Successful logon (T1078), Privilege escalation (T1548.002), and Egress (T1071.001). Three pieces of evidence are linked, each with a weight (w=0.94, w=0.91, w=0.88).

CRITICAL **ACTIVE** ALT-29FE6224F202

Privilege escalation to SYSTEM (+2 related from 3 tools)

ATTACK STORYBOARD · 4 STAGES

RISK SCORE
97

CONFIDENCE
93%

FIRST SEEN
20:54

LAST SEEN
20:54

Mitigate Attack

ATTACK CHAIN

- 11:40 Brute force** (T1110)
Repeated failed logons against mmalek
- 11:44 Successful logon** (T1078)
mmalek authenticated to HR-WIN-11
- 11:49 Privilege escalation** (T1548.002)
Escalation to SYSTEM via UAC bypass
- 11:54 Egress** (T1071.001)
Outbound connection to 45.9.148.117

EVIDENCE LINKED

- EV-CORR-1 AUTH** (w=0.94)
Successful logon for mmalek after repeated failures
src: HR-WIN-11
- EV-CORR-2 PROCESS** (w=0.91)
powershell.exe escalated to SYSTEM
src: HR-WIN-11
- EV-CORR-3 NETWORK** (w=0.88)
Egress to 45.9.148.117
src: 10.9.4.7

- The header row shows the **risk score**, **confidence**, **first seen** and **last seen**.
- **Attack chain**. Each stage of the attack in time order, with its MITRE ATT&CK technique id (for example T1110 Brute force).
- **Evidence linked**. The individual pieces of evidence, their type (auth, process, network) and weight.
- **Mitigate Attack** (red button). Marks the incident **CONTAINED** by hand, **without** running the plan. Use it only when the threat was already contained outside AO-SOC. The normal route is **Approve plan** (Section 6).

Security Situation: why several alerts became one incident

Security Situation

3 DETECTION(S) CORRELATED FROM 3 TOOL(S)

EDGE-FIREWALL SPLUNK WAZUH Independently corroborated

CORRELATION RISK SCORE

100/100

- +88 Highest severity among 3 detection(s) is CRITICAL
- +24 Independently detected by 3 tools: edge-firewall, splunk, wazuh
- +4 3 detections inside the correlation window
- +5 3 distinct techniques asserted by the detecting tools: T1071.001, T1110, T1548.002

CORRELATED ON

mmalek 203.0.113.44 18.9.4.7 45.9.148.117 HR-WIN-11 powershell.exe

MEMBER DETECTIONS

> SPLUNK MEDIUM 07:10

Brute force against a single account

> WAZUH CRITICAL 07:19

Privilege escalation to SYSTEM

> EDGE-FIREWALL HIGH 07:24

Outbound connection to a low-reputation host

- **Tool badges** (for example EDGE-FIREWALL , SPLUNK , WAZUH) name the tools that reported it. **Independently corroborated** means more than one tool saw it.
- **Correlation risk score.** Shows the arithmetic, line by line: the points for the highest severity, for cross-tool corroboration, for the number of detections and for the number of techniques. The score is a count of facts, not the AI's opinion.
- **Correlated on.** The shared users, addresses, hosts and processes that joined the detections.
- **Member detections.** Every original detection. Click > to expand one and see its entities, techniques and a link back to the tool that raised it.

Verification and precedent

Verification and precedent

WHAT WAS CHECKED AGAINST A SOURCE OF RECORD, AND WHAT THIS SOC DECIDED BEFORE

MALICIOUS 45.9.148.117 sample-indicator-set credential-theft egress sample-data

2 not checked (internal addresses and identities are never sent to a feed)

PRECEDENT

No sufficiently similar past decision — this one waits for an analyst by construction.

- **Indicators checked against threat intelligence.** MALICIOUS means the indicator is on a feed. *Checked and not found* is **not** proof that it is safe. *Not checked* covers internal addresses and identities, which are

never sent to a feed.

- **Precedent.** What this SOC decided in similar past cases. "*No sufficiently similar past decision*" means the incident waits for a person by design.

A quiet panel is not a clean result. If no intelligence feed is configured, the panel says so.

Case: who owns this incident

Case
WHO IS WORKING THIS, AND WHAT STATE IT IS IN

CASE-2226E77515 **New** **P1**

Nobody owns this case yet

Assign to... [Return to the queue](#)

Assigned **In progress** **Escalated** **Resolved** **Closed** **Escalate a tier**

Add a note to the case file... [Add note](#)

System of record
No ticketing system is configured. This case lives here only — it is not out of sync, there is nothing to be in sync with.

Case file [Refresh](#)

- Case opened for situation SIT-EB230A5A206B — ao-soc · 9/21/2026, 8:54:17 PM

- The **case id**, its **state** and its **priority** (**P1** is the highest).
- **Assign to...** Type a name and click **Assign**. **Return to the queue** unassigns it.
- **State buttons** move the case through its lifecycle: **New** → **Assigned** → **In progress** → **Escalated** → **Resolved** → **Closed** , and **Reopened** from **Resolved** or **Closed**. Only the valid next states are shown.
- **Escalate a tier** raises the case to the next tier.
- **Add a note** writes to the **Case file**, the case's timeline, which cannot be edited afterwards.
- **System of record.** Shows whether the case is synced to a ticketing system (for example TheHive). "*No ticketing system is configured*" is a normal state, not an error.

Nothing in the Case panel can approve or reject a decision. Owning a case and deciding what to do about it are deliberately separate.

AI Recommended Actions

 **AI Recommended Actions** 3 ACTIONS

 **Disable account** A1

Target: **mmalek**

Reason

Confirmed credential compromise

Impact

Locks the user out

Confidence

94%

Approve the Tier-2 plan above to auto-execute

 **Isolate host** A2

Target: **HR-WIN-11**

Reason

SYSTEM-level compromise

Impact

Contains the endpoint

Confidence

92%

Approve the Tier-2 plan above to auto-execute

 **Block IP** A3

Target: **45.9.148.117**

Reason

Low-reputation egress peer

Impact

Stops egress

Confidence

98%

Approve the Tier-2 plan above to auto-execute

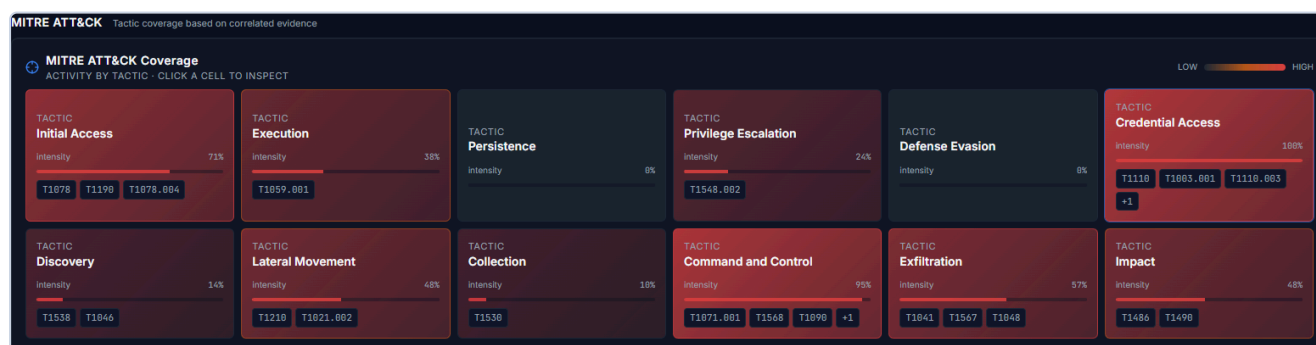
The same actions as the plan, each with its **target**, **reason**, **impact** and **confidence**. For live incidents you do not run these one by one. The note "*Approve the Tier-2 plan above to auto-execute*" tells you to use **Approve plan**.

5.3 Risk Analytics



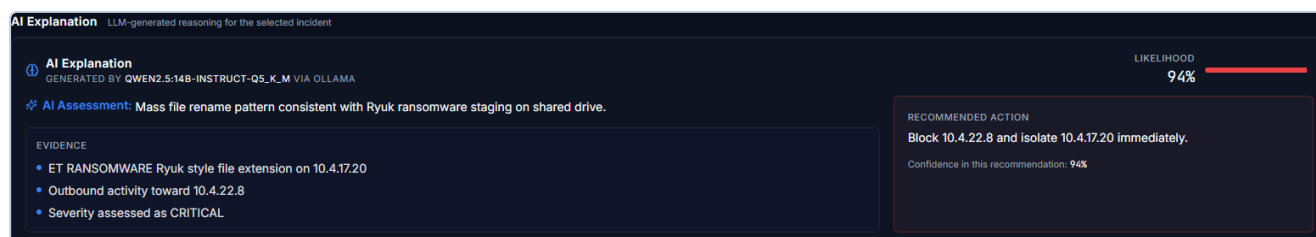
The highest-risk **users**, **hosts** and **IP addresses** driving the current posture. The **Entity Risk** page (Section 10) has the full, searchable list.

5.4 MITRE ATT&CK



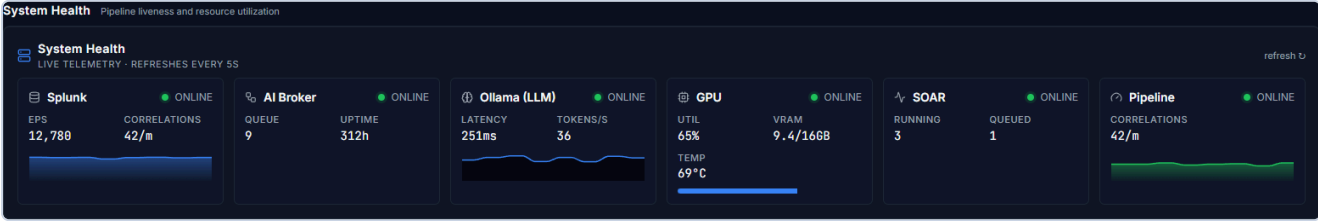
Activity by ATT&CK **tactic**. Redder cells mean more intensity. Each cell lists the technique ids seen. Click a cell to inspect it. Techniques are checked against a local ATT&CK catalogue, and an id the catalogue does not list is flagged as *unlisted* rather than presented as fact.

5.5 AI Explanation



The local model's reasoning for the selected incident: its **assessment**, the **evidence** it relied on, its **recommended action** and its **likelihood**. The panel also names the model that wrote it (for example qwen2.5:14b-instruct via Ollama).

5.6 System Health (panel)



A compact version of the System Health page (Section 11).

6. Working an incident: the AI Tier-2 Decision

This panel is the heart of AO-SOC. You **review once**, and the response tools then run the whole plan.

AI Tier-2
Decision

REVIEW ONCE — SOAR RUNS THE FULL
PLAN AUTOMATICALLY

CONTAIN

Pending approval

AI verdict

Confidence: 93%

RATIONALE

Three unrelated tools independently observed stages of the same intrusion against one account and one host. No single alert justifies containment; the correlation does.

Risk if executed

Disabling mmalek locks an HR user out mid-shift and isolating HR-WIN-11 drops any in-flight payroll session on that endpoint.

4

UNDLED ACTION PLAN (3)

Disable account

High-risk write

PENDING

Target: mmalek

Confirmed credential compromise

Can be undone

Undo with: Enable account

Rollback becomes available once the action has run

Isolate host

High-risk write

PENDING

Target: HR-WIN-11

SYSTEM-level compromise

Can be undone

Undo with: Release host from isolation

Rollback becomes available once the action has run

Block IP

High-risk write

PENDING

Target: 45.9.148.117

Low-reputation egress peer

Can be undone

Undo with: Unblock IP

Rollback becomes available once the action has run

2

Reject

3

Edit

1

Approve plan

5

AUDIT TRAIL

Reading the panel from top to bottom

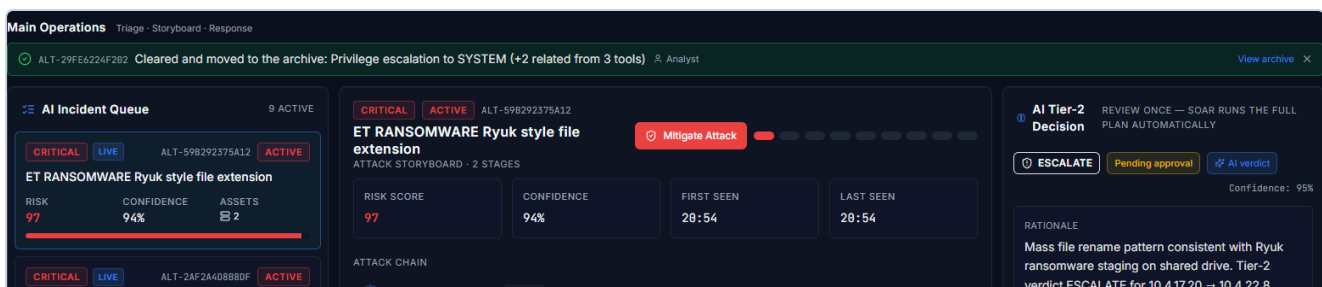
- **Verdict:** CONTAIN , ESCALATE , INVESTIGATE , MONITOR OR IGNORE .

- **Status:** Pending approval → Approved → Executing → Complete . Also possible: Rejected , Failed , Dry run , Superseded .
- **Who decided:**
 - **AI verdict:** the model produced the decision.
 - **Rule fallback:** the model gave no usable answer, so a severity rule decided. Read these more carefully.
 - **Analyst corrected:** a person edited it.
- **Rationale:** why, in plain language.
- **Risk if executed** (amber box): what the plan could break. For example, disabling an HR user in the middle of their shift.
- **(4) Bundled action plan:** each action with:
 - its **target** (IP address, host or account) and reason,
 - its **risk class:** Read , Low-risk write , High-risk write or Destructive ,
 - its **reversibility:** Can be undone (with the undo action shown), Cannot be undone, or Nothing to undo,
 - a guard badge where it applies: **Critical asset**, **Privileged account** or **Service account**. Autopilot never runs these; a person must approve them.
- **Buttons: (1) Approve plan, (2) Reject, (3) Edit.**
- **(5) Audit trail:** how the decision was produced (Section 6.4).

6.1 Approve the plan

1. Read the **rationale**, the **Security Situation** and the **Risk if executed** box.
2. Check every **target** in the plan. Is it the right host, account or address?
3. Click **Approve plan**.

Every action is queued and runs automatically. The status of each action changes from **PENDING** to **EXECUTING** and then **DONE** . When they all succeed, the incident is **cleared**: it leaves the queue, moves to the **Archive**, and a green banner announces it.



Click **View archive** to see it, or **X** to dismiss the banner.

Dry run. When the deployment runs with `RESPONSE_DRY_RUN=true` , actions show **Dry run** (amber) instead of **Complete**: *"nothing was sent to any executor"*. The incident is **not** marked contained. This is deliberate, because a simulated containment must never look like a real one.

6.2 Edit the plan (correct the AI)

Use **Edit** when the AI is *partly* right: the verdict is wrong, a target is wrong, or an action is missing or unnecessary.

AI Tier-2 Decision

REVIEW ONCE — SOAR RUNS THE FULL PLAN AUTOMATICALLY

CONTAIN

Pending approval

AI verdict

Confidence: 93%

RATIONALE

Three unrelated tools independently observed stages of the same intrusion against one account and one host. No single alert justifies containment; the correlation does.

Risk if executed

Disabling mmalek locks an HR user out mid-shift and isolating HR-WIN-11 drops any in-flight payroll session on that endpoint.

BUNDLED ACTION PLAN (3)

Disable account

High-risk write

PENDING

Target: mmalek

Confirmed credential compromise

Can be undone

Undo with: Enable account

Rollback becomes available once the action has run

Isolate host

High-risk write

PENDING

Target: HR-WIN-11

SYSTEM-level compromise

Can be undone

Undo with: Release host from isolation

Rollback becomes available once the action has run

Block IP

High-risk write

PENDING

Target: 45.9.148.117

Low-reputation egress peer

Can be undone

Undo with: Unblock IP

Rollback becomes available once the action has run

CORRECT THE PLAN

Approve or reject records that the machine was wrong. Editing records what right looks like — the change is kept as a label the system learns from.

Verdict

CONTAIN

Actions

Disable account

mmalek

Confirmed credential compromise

Isolate host

HR-WIN-11

SYSTEM-level compromise

Block IP

45.9.148.117

Low-reputation egress peer

+ Add action

What the machine got wrong (optional)

e.g. read the severity, missed the approved change ticket

Cancel

Save correction

> AUDIT TRAIL

1. Click **Edit**.
2. Change the **Verdict** if needed.

3. Change any action's name, target or reason. **Add action** adds a row, and the bin icon removes one.
4. Optionally, write **what the machine got wrong**, for example "*missed the approved change ticket*".
5. Click **Save correction**, then approve the corrected plan.

Your correction is stored as a **label** that the system learns from. It feeds the *precedent* that later decisions are compared with. Editing is how the system gets better, so an edit is more useful than a silent reject.

AO-SOC refuses an edit that could never run, for example a *Block IP* action whose target is not an IP address. A plan that has already been executed cannot be edited, because it is the record of what was sent.

6.3 Reject the plan

AI Tier-2 Decision

REVIEW ONCE — SOAR RUNS THE FULL PLAN AUTOMATICALLY

CONTAIN

Pending approval

AI verdict

Confidence: 93%

RATIONALE

Three unrelated tools independently observed stages of the same intrusion against one account and one host. No single alert justifies containment; the correlation does.

Risk if executed

Disabling mmalek locks an HR user out mid-shift and isolating HR-WIN-11 drops any in-flight payroll session on that endpoint.

BUNDLED ACTION PLAN (3)

Disable account

High-risk write

PENDING

Target: mmalek

Confirmed credential compromise

Can be undone

Undo with: Enable account

Rollback becomes available once the action has run

Isolate host

High-risk write

PENDING

Target: HR-WIN-11

SYSTEM-level compromise

Can be undone

Undo with: Release host from isolation

Rollback becomes available once the action has run

Block IP

High-risk write

PENDING

Target: 45.9.148.117

Low-reputation egress peer

Can be undone

Undo with: Unblock IP

Rollback becomes available once the action has run

Rejection note (optional)

Why is this plan being rejected?

Cancel

Reject

AUDIT TRAIL

1. Click **Reject**.
2. Optionally, write **why** in the rejection note.
3. Confirm with **Reject**.

Nothing runs. Use Reject when no action is warranted, for example for an authorised scanner or a known false positive.

6.4 Audit trail

AI Tier-2 Decision

REVIEW ONCE — SOAR RUNS THE FULL PLAN AUTOMATICALLY

CONTAIN

Pending approval

AI verdict

Confidence: 93%

RATIONALE

Three unrelated tools independently observed stages of the same intrusion against one account and one host. No single alert justifies containment; the correlation does.

Risk if executed

Disabling mmalek locks an HR user out mid-shift and isolating HR-WIN-11 drops any in-flight payroll session on that endpoint.

BUNDLED ACTION PLAN (3)

Disable account

High-risk write

PENDING

Target: mmalek

Confirmed credential compromise

Can be undone

Undo with: Enable account

Rollback becomes available once the action has run

Isolate host

High-risk write

PENDING

Target: HR-WIN-11

SYSTEM-level compromise

Can be undone

Undo with: Release host from isolation

Rollback becomes available once the action has run

Block IP

High-risk write

PENDING

Target: 45.9.148.117

Low-reputation egress peer

Can be undone

Undo with: Unblock IP

Rollback becomes available once the action has run

Reject

Edit

Approve plan

AUDIT TRAIL

Autonomy

Proposed — nobody has approved it

Model

scripted · scripted

Run

RUN-ACB45FAE43D6

Reasoning hash

0288a6be689d...dc488f45

Integrity

Verified

Confidence is the model's own report and is not calibrated — It is shown, never used to decide.

Download JSON

Click **Audit trail** to expand it. It shows:

- **Autonomy:** *Proposed* (nobody has approved it yet), *Supervised* (a person approved it) or *Autonomous* (autopilot approved it on precedent).
- **Model** and **Run:** which model produced the decision, and the id of that run.
- **Reasoning hash** (copy button) and **Integrity:** `Verified` means the stored prompt and answer still match their hash. `Mismatch` means the record was altered.
- **Download JSON:** the complete decision envelope (situation, decision, actions and audit trail), for reports or evidence.

6.5 Record what actually happened

After a decision settles, the panel asks **What actually happened**, and you can answer for 72 hours (configurable):

- **True positive:** it was a real attack.
- **False positive:** it was not.
- **Reopened:** it came back.

Please record an outcome. These answers decide which detection sources and verdicts are trustworthy enough ever to be automated.

6.6 Autopilot (if enabled)

When a site turns autopilot on, some decisions run **without** a click. Autopilot does this only when **enough similar past cases were confirmed by people with the same verdict, none were reversed, and the newest is recent**. It **never** automates:

- actions on a **critical asset** (domain controllers, databases and anything else the site lists),
- disabling or resetting a **privileged** or **service** account,
- an action that **cannot be undone**,
- a **destructive** action.

In the Archive, incidents that autopilot cleared carry an **Autopilot** badge instead of **Analyst**.

7. Live Alerts and the Incident Playbook

Menu: Live Alerts. The raw alert log from the broker, for fast scanning.

AO-SOC
COMMAND
CENTER

Command
Center

Live
Alerts

Incidents

Archive

Entity
Risk

System
Health

Splunk

AI
Broker

LLM

SOAR

POSTURE **ELEVATED** (97)

EN

08:25:01
SEP 22, 2028 · GMT+3:30

Live Alert Log

Splunk-ingested alerts from the Aegis-Link broker

Refresh

LIVE ALERTS
13

PENDING
9

CONTAINED
4

CRITICAL
2

HIGH
5

MEDIUM / LOW
2

Alert Log

9 BROKER ALERTS

TIME	SEVERITY	STATUS	SOURCE	DESTINATION	SIGNATURE
20:54	CRITICAL	ACTIVE	10.4.17.20	10.4.22.8	ET RANSOMWARE Ry...
20:54	CRITICAL	ACTIVE	10.4.9.17	10.4.21.50	ET EXPLOIT Possible ...
20:54	HIGH	ACTIVE	10.4.73.16	198.51.100.14	ET INFO Suspicious T...
20:54	HIGH	ACTIVE	10.4.59.14	8.8.8.8	ET DNS Query for .onl...
20:54	HIGH	ACTIVE	10.4.90.12	193.142.146.4	ET POLICY Suspicious...
20:54	HIGH	ACTIVE	10.4.103.18	203.8.113.44	ET POLICY Powershel...
20:54	HIGH	ACTIVE	10.4.21.10	185.220.101.7	ET MALWARE Known ...
20:54	MEDIUM	ACTIVE	10.4.38.15	177.185.83.40	ET WEB_SERVER SQL ...
20:54	MEDIUM	ACTIVE	10.4.45.11	45.155.205.211	ET SCAN Potential SS...

Incident Playbook

ALT-590292375A12

Mitigate Attack

AI ANALYSIS

Mass file rename pattern consistent with Ryuk ransomware staging on shared drive.

RECOMMENDED CONTAINMENT

Block egress to 10.4.22.8 at the perimeter firewall

Isolate host 10.4.17.20 from the network segment

Collect memory dump and triage for credential theft

PRIMARY RECOMMENDATION

Block 10.4.22.8 and isolate 10.4.17.20 immediately.

AO-SOC Command Center · v2.8.7

Splunk · AI Broker · Local LLM (Qwen) · SOAR

- **Top tiles:** counts of **Live alerts**, **Pending**, **Contained**, **Critical**, **High** and **Medium / Low**.
- **Alert Log:** time, severity, status, source and destination addresses, and the signature. **Refresh** reloads the log.

Click an alert to open its **Incident Playbook** on the right:

AO-SOC
COMMAND
CENTER

Command Center

Live Alerts

Incidents

Archive

Entity Risk

System Health

Splunk

AI Broker

LLM

SOAR

POSTURE ELEVATED (97)

EN

08:25:83
SEP 22, 2028 · GMT+3:30

Live Alert Log

Splunk-ingested alerts from the Aegis-Link broker

Refresh

LIVE ALERTS 13

PENDING 9

CONTAINED 4

CRITICAL 2

HIGH 5

MEDIUM / LOW 2

Alert Log

9 BROKER ALERTS

TIME	SEVERITY	STATUS	SOURCE	DESTINATION	SIGNATURE
28:54	CRITICAL	ACTIVE	10.4.17.28	10.4.22.8	ET RANSOMWARE Ry...
28:54	CRITICAL	ACTIVE	10.4.9.17	10.4.21.58	ET EXPLOIT Possible ...
28:54	HIGH	ACTIVE	10.4.73.16	198.51.188.14	ET INFO Suspicious T...
28:54	HIGH	ACTIVE	10.4.59.14	8.8.8.8	ET DNS Query for .onl...
28:54	HIGH	ACTIVE	10.4.98.12	193.142.146.4	ET POLICY Suspicious...
28:54	HIGH	ACTIVE	10.4.183.18	203.0.113.44	ET POLICY Powershel...
28:54	HIGH	ACTIVE	10.4.21.10	185.228.101.7	ET MALWARE Known ...
28:54	MEDIUM	ACTIVE	10.4.38.15	177.185.83.48	ET WEB_SERVER SQL ...
28:54	MEDIUM	ACTIVE	10.4.45.11	45.155.285.211	ET SCAN Potential SS...

Incident Playbook

ALT-2AF2A4D8B8DF

Mitigate Attack

AI ANALYSIS

SMB exploit traffic targeting file server — immediate lateral movement risk.

RECOMMENDED CONTAINMENT

Block egress to 10.4.21.50 at the perimeter firewall

Isolate host 10.4.9.17 from the network segment

Collect memory dump and triage for credential theft

PRIMARY RECOMMENDATION

Block 10.4.21.50 and isolate 10.4.9.17 immediately.

AO-SOC Command Center - v2.8.7

Splunk → AI Broker → Local LLM (Qwen) → SOAR

- **AI analysis:** a one-line summary.
- **Recommended containment:** the playbook steps as a checklist. A step turns green with a tick once it has been done.
- **Primary recommendation:** the single most important step.
- **Mitigate Attack:** marks the alert contained by hand (see Section 5.2).

To **approve and run** the playbook, open the incident in the **Command Center** or on its **details page** and use **Approve plan** (Section 6).

8. Incidents list and Incident details

8.1 Incidents

Menu: Incidents. Every active incident in one table.

AO-SOC

COMMAND CENTER

Command Center

Live Alerts

Incidents

Archive

Entity Risk

System Health

Splunk

AI Broker

LLM

SOAR

EN

FA

08:25:05

SEP 22, 2026 - GMT+3:30

Incidents

All correlated incidents across the environment

Incident Queue

13 TOTAL

SEVERITY		STATUS	TITLE	RISK	CONF	OPEN
CRITICAL	LIVE	ACTIVE	ALT-59B292375A12 ET RANSOMWARE Ryuk style file extension	97	94%	→
CRITICAL	LIVE	ACTIVE	ALT-2AF2A40888DF ET EXPLOIT Possible EternalBlue SMB attempt	97	94%	→
CRITICAL	DEMO	ACTIVE	INC-2B41 Possible Ransomware Activity on FIN-WIN-04	97	94%	→
HIGH	LIVE	ACTIVE	ALT-7C421F761BCC ET INFO Suspicious TLS SNI to DGA-like domain	78	88%	→
HIGH	LIVE	ACTIVE	ALT-F8C488E86C8B ET DNS Query for .onion TLD	78	88%	→
HIGH	LIVE	ACTIVE	ALT-2DC746149A2F ET POLICY Suspicious inbound to MSSQL port 1433	78	88%	→
HIGH	LIVE	ACTIVE	ALT-7311B3170BF4 ET POLICY Powershell DownloadString	78	88%	→
HIGH	LIVE	ACTIVE	ALT-811AE0AA66ED ET MALWARE Known C2 Beacon	78	88%	→
HIGH	DEMO	INVESTIGATING	INC-2B4B Brute Force Campaign against VPN Gateway	78	88%	→
MEDIUM	LIVE	ACTIVE	ALT-FEC2C8E32686 ET WEB_SERVER SQL Injection Attempt	62	71%	→
MEDIUM	LIVE	ACTIVE	ALT-Z7B5C548DA65 ET SCAN Potential SSH Scan	62	71%	→
MEDIUM	DEMO	OPEN	INC-2B39 Suspicious Cloud IAM Role Assumption	62	71%	→
LOW	DEMO	MONITORING	INC-2B38 Outdated TLS Handshake on External Endpoint	31	64%	→

AO-SOC Command Center - v2.8.7

Splunk + AI Broker + Local LLM (Qwen) + SOAR

The columns are severity, status (with a `LIVE` or `DEMO` badge), title, **risk**, AI **confidence**, and **Open**. Click a row to open the incident's details page.

8.2 Incident details

The screenshot displays the AO-SOC Command Center interface for a specific incident. The top navigation bar includes links to Command Center, Live Alerts, Incidents, Archive, Entity Risk, and System Health. The incident title is "ET RANSOMWARE Ryuk style file extension" with a severity of "CRITICAL" and status of "ACTIVE". The left sidebar contains several sections: "Attack Storyboard" showing a timeline of events, "Security Situation" with a risk score of 45/100, "Verification and precedent" section, and "Case" section. The right sidebar features the "AI Tier-2 Decision" section, which includes an "ESCALATE" button, a "Rationale" section, a "Risk if executed" warning, and a "BUNDLED ACTION PLAN" with items like "Block IP", "Isolate Host", and "Investigate". At the bottom right, there are buttons for "Reject", "Edit", and "Approve plan", and an "AUDIT TRAIL" link.

One page per incident, with everything in one place:

- **Back to Command Center** link, the incident id, severity, status and title.
- **Left column:** the **Attack Storyboard**, **Security Situation**, **Verification and precedent**, **Case** and **AI Explanation**.
- **Right column:** the **AI Tier-2 Decision** (with all its buttons), **Affected Assets**, **MITRE Techniques** and **AI Recommended Actions**.

Use this page when you want to share a link to one incident, or to **roll back** an action on an incident that is already contained (Section 9.2).

9. Archive and rolling actions back

9.1 Archive

Menu: Archive. Cleared incidents, newest first.

The screenshot shows the 'Incident Archive' section of the AO-SOC Command Center. The top navigation bar includes 'Command Center', 'Live Alerts', 'Incidents', 'Archive' (selected), 'Entity Risk', and 'System Health'. The 'Archive' tab shows a list of cleared incidents. The first incident is titled 'Privilege escalation to SYSTEM (+2 related from 3 tools)' with a severity of 'CRITICAL' and a status of 'CONTAINED'. It includes a 'Rationale' section stating that three unrelated tools independently observed stages of the same intrusion. Below the rationale, three SOAR actions are listed: 'Disable account', 'Isolate host', and 'Block IP', each with a status of 'DONE' and an execution ID. The incident is approved by 'analyst' with a confidence of 93% and decided by 'AI verdict'. The second incident is titled 'ET TROJAN Possible Zeus variant outbound' with a severity of 'CRITICAL' and a status of 'CONTAINED'. The third incident is titled 'ET SCAN NMAP -sS window 1024' with a severity of 'LOW' and a status of 'CONTAINED'. The fourth incident is titled 'ET POLICY Outbound SMTP to suspicious port' with a severity of 'MEDIUM' and a status of 'CONTAINED'. The bottom of the screen shows the version 'AO-SOC Command Center - v2.8.7' and the system components 'Splunk + AI Broker + Local LLM (Qwen) + SOAR'.

- The header counts the incidents **cleared** and how many were **auto-executed**.
- Each row shows the severity, **CONTAINED**, the verdict, and **Analyst** or **Autopilot**: who approved it.
- **Click a row** to expand it and see the **rationale**, every **delivered SOAR action** with its status and execution id, who approved it, its confidence and its decision source.
- **Open incident details** opens the full incident page.

9.2 Roll back an action

Any action marked **Can be undone** can be reversed after it has run. For example, a *Block IP* is undone with *Unblock IP*.

1. From the Archive, click **Open incident details**. You can also use the Command Center if the incident is still active.
2. In the AI Tier-2 Decision panel, find the action and click **Roll back**.

AI Tier-2 Decision

REVIEW ONCE — SOAR RUNS THE FULL PLAN AUTOMATICALLY

CONTAIN

Complete

AI verdict

Confidence: 93%

RATIONALE

Three unrelated tools independently observed stages of the same intrusion against one account and one host. No single alert justifies containment; the correlation does.

Risk if executed

Disabling mmalek locks an HR user out mid-shift and isolating HR-WIN-11 drops any in-flight payroll session on that endpoint.

BUNDLED ACTION PLAN (3)

Disable account

High-risk write

DONE

Target: mmalek

Confirmed credential compromise

via soar

exec_22be47ba6f

Can be undone

Undo with: Enable account

1 Roll back

Isolate host

High-risk write

DONE

Target: HR-WIN-11

SYSTEM-level compromise

via soar

exec_e7f5edd296

Can be undone

Undo with: Release host from isolation

Roll back

Block IP

High-risk write

DONE

Target: 45.9.148.117

Low-reputation egress peer

via soar

exec_b7aa51e6ba

Can be undone

Undo with: Unblock IP

Roll back

SOAR execution complete — incident contained

WHAT ACTUALLY HAPPENED

Reportable for 72h after the decision settles.

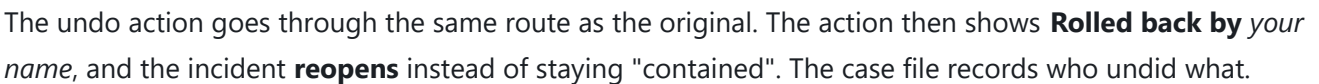
True positive

False positive

Reopened

AUDIT TRAIL

- Optionally, write **why**, for example "*change window CHG-2211 approved this traffic*".
- Click **Roll back** again to confirm.

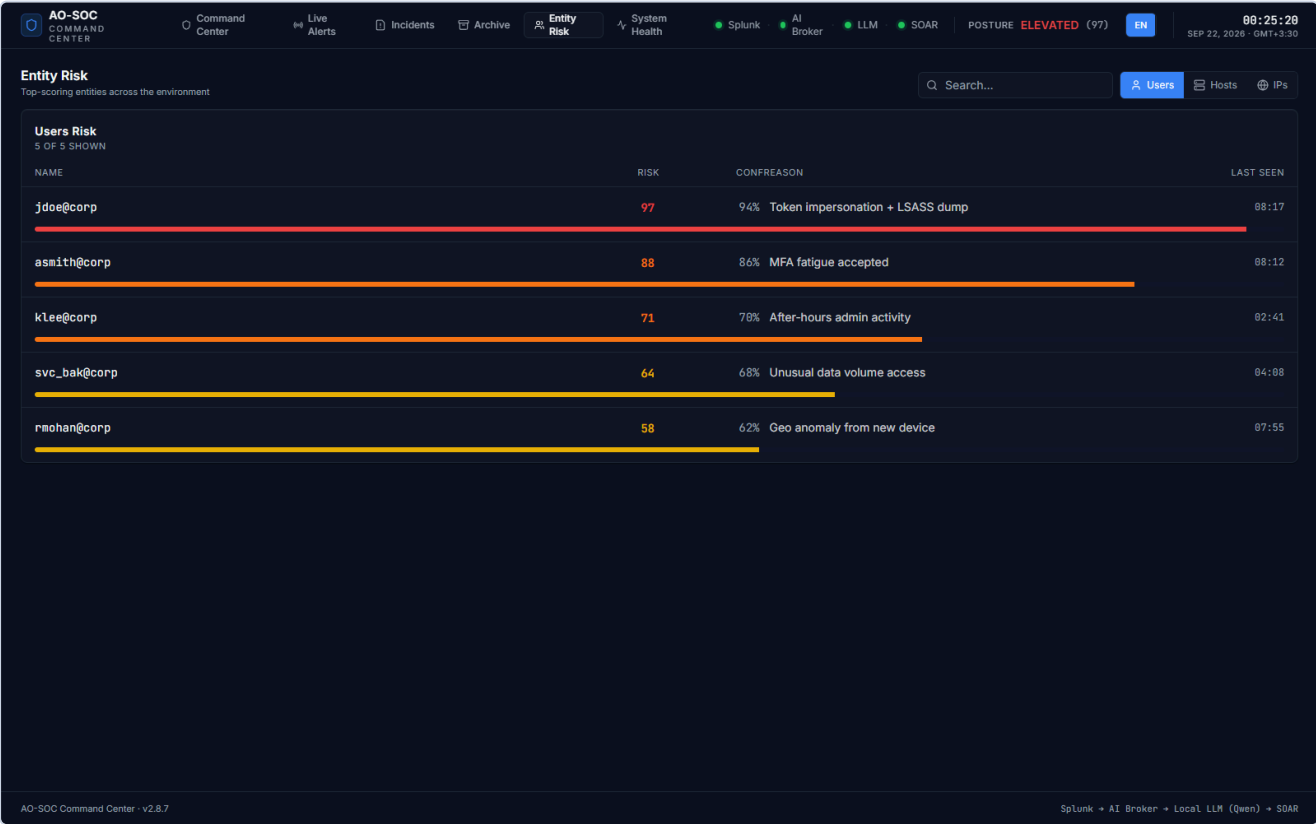


- AO-SOC Command Center — User Manual · v1.0 · © J.Ekrami-Labs

- "*The executor refused the rollback*" means the response tool has no undo command configured. Undo the action in that tool directly.

10. Entity Risk

Menu: Entity Risk. The highest-scoring entities across the environment.

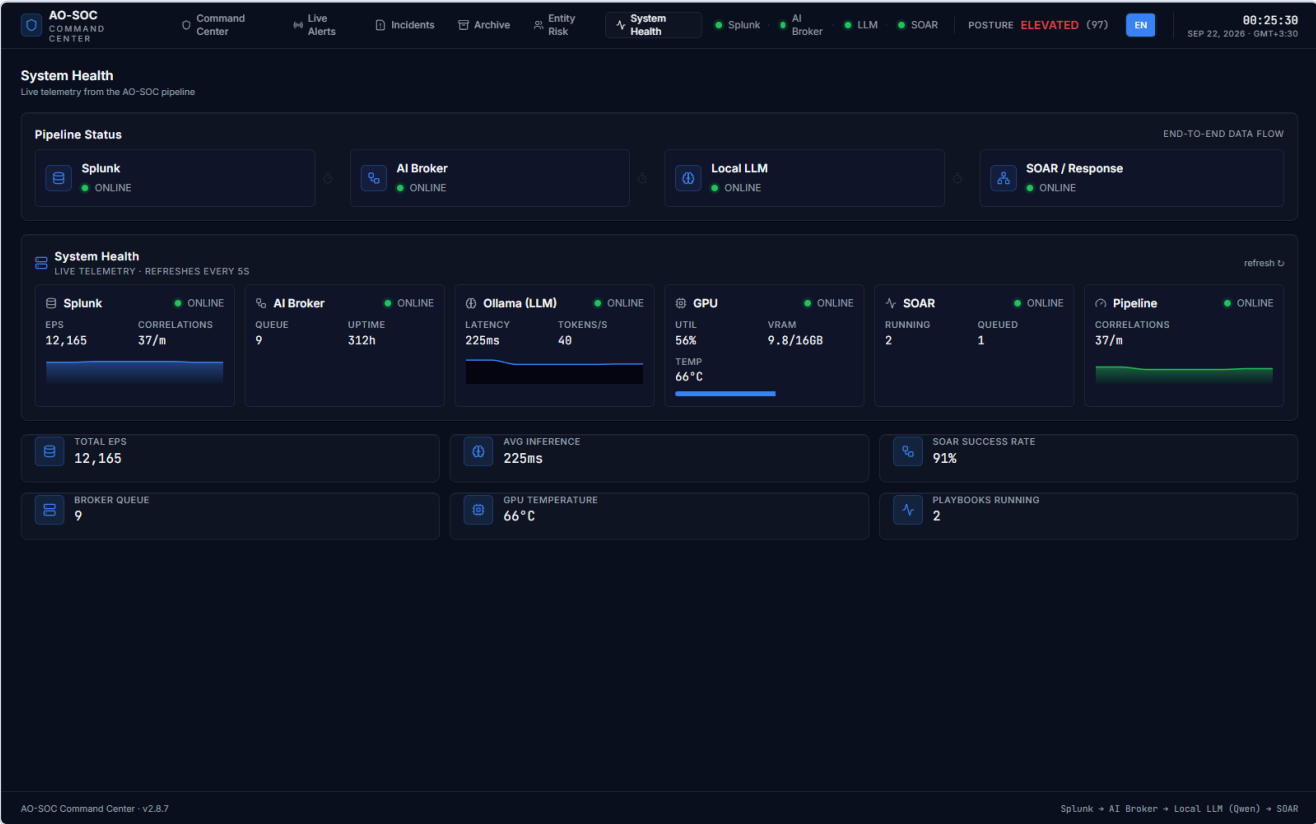


- Use the tabs on the right to switch between **Users**, **Hosts** and **IPs**.
- **Search...** filters the list as you type.
- Each row shows the name, the **risk** score (with a coloured bar), the model's **confidence**, the **reason** and when it was **last seen**.

Use this page to answer "*who or what is most at risk right now?*", and to check whether an account or host from an incident already has a history.

11. System Health

Menu: **System Health**. Is the pipeline alive?

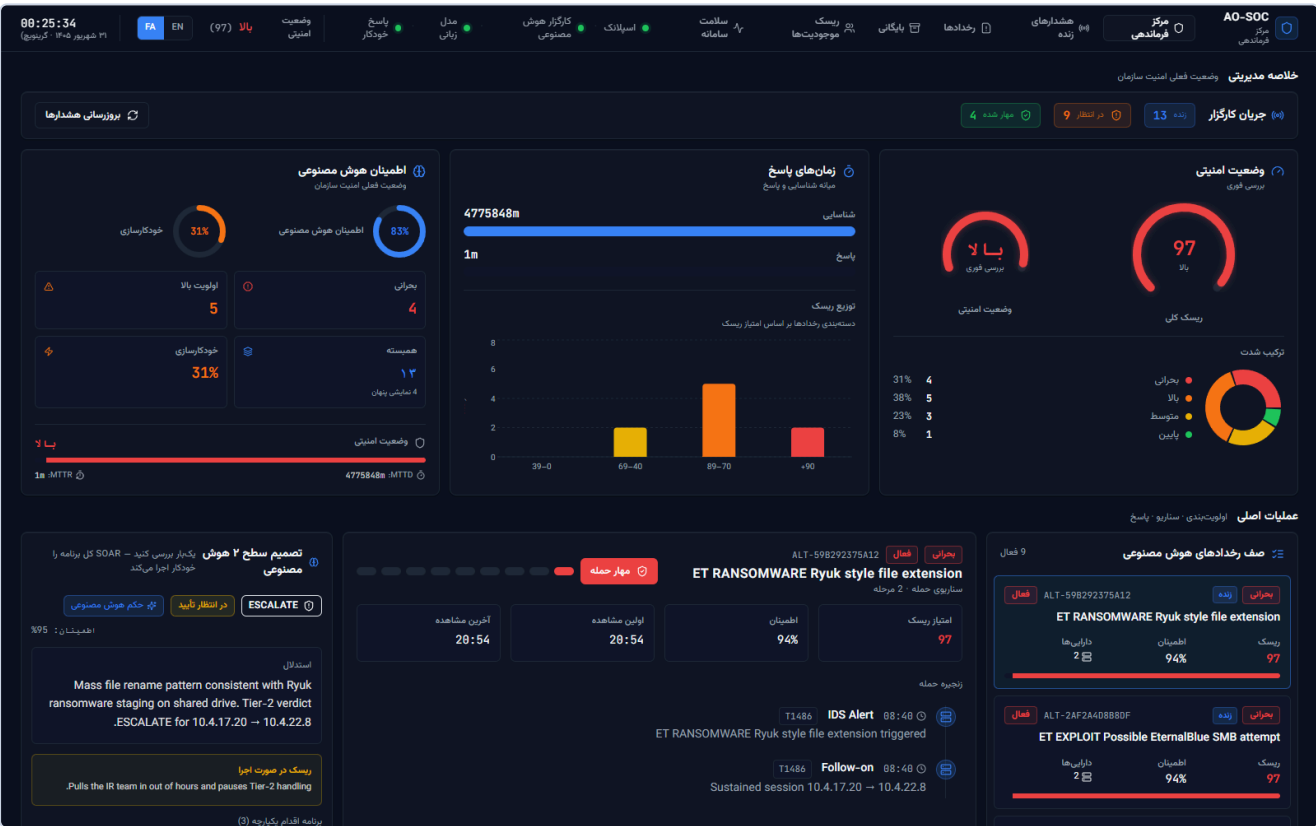


- **Pipeline Status:** **Splunk** → **AI Broker** → **Local LLM** → **SOAR / Response**, each **ONLINE** , **DEGRADED** , **OFFLINE** or **UNKNOWN** .
- **Live telemetry** (refreshed every 5 seconds): events per second and correlations (Splunk), queue depth and uptime (AI Broker), latency and tokens per second (the model), GPU utilisation, VRAM and temperature, and SOAR actions running and queued.
- **Summary tiles:** total EPS, average inference time, SOAR success rate, broker queue, GPU temperature and playbooks running.

If a component is red or grey, see [Section 15](#).

12. Language: English and Persian

Click **FA** in the top bar to switch to Persian, and **EN** to switch back. The whole interface changes to right-to-left, including labels, dates and risk explanations. The choice is remembered in this browser.



13. A daily routine

At the start of a shift

1. Sign in with **your own** key.
2. Look at the top bar: are all four pipeline dots **green**? If not, open **System Health**.
3. Read the **Executive Summary**: the posture, how many incidents are **PENDING**, and the severity mix.

During the shift, for each incident in the AI Incident Queue (top first)

1. **Select it**.
2. **Read**: the storyboard, then the **Security Situation** (what joined it together), then **Verification and precedent**.
3. **Take ownership**: in **Case**, assign it to yourself and set it to *In progress*.
4. **Decide**, in the AI Tier-2 Decision panel:
 - the plan is right → **Approve plan**,
 - the plan is partly wrong → **Edit**, then approve,
 - nothing should run → **Reject**, with a note,
 - it needs a higher tier → **Escalate a tier** in the Case panel.
5. **Note** anything useful in the case file.
6. When it clears, check the **Archive** entry.

At the end of the shift

1. For decisions that settled today, record **True positive / False positive / Reopened**.
2. Hand over any case that is still *In progress*: add a note and reassign it.
3. Close the browser tab. This signs you out.

For the shift lead (weekly): take and verify a backup (see the runbook, section 7), and review which detection sources produce false positives.

14. Glossary: badges, colours and statuses

Severity colours: CRITICAL red · HIGH orange · MEDIUM yellow · LOW green.

Posture: LOW → GUARDED → HEIGHTENED → ELEVATED (the worst).

Term	Meaning
LIVE / DEMO	Real data from the broker / built-in sample data
Situation	One or more detections, possibly from several tools, about the same entities within a time window
Tier-2 decision	The verdict: CONTAIN (act now), ESCALATE (bring in a higher tier or IR), INVESTIGATE (look deeper), MONITOR (watch), IGNORE (no action)
Bundled action plan	The playbook: the set of actions attached to a decision, approved and run together
AI verdict / Rule fallback / Analyst corrected	Whether the model, a severity rule or a person produced the decision
Read / Low-risk write / High-risk write / Destructive	How much an action changes. Destructive actions are refused unless a site enables them deliberately
Can be undone / Cannot be undone / Nothing to undo	Whether the action can be rolled back
Critical asset / Privileged account / Service account	Targets autopilot never touches
Pending approval → Approved → Executing → Complete	The normal status sequence of a plan
Failed	The response tool answered and declined. The reason is in the receipt
Blocked	The action never left AO-SOC. The target was malformed, or no tool performs it
Dry run	Simulated only. Nothing was sent to any tool
Superseded	The situation was merged into another one, and the live decision is there
Precedent	Similar past cases that people decided, which autopilot relies on
Case states	New , Assigned , In progress , Escalated , Resolved , Closed , Reopened
MTTD / MTTR	Mean time to detect / mean time to respond

15. Troubleshooting

Symptom	What to do
The sign-in screen keeps coming back	The key is wrong, revoked, or belongs to the broker rather than the dashboard. Ask your administrator for a key from <code>AOSOC_API_KEYS</code>
"Failed to load dashboard ... port 4317"	The UI API is not running. Demo: restart with <code>start-demo</code> . Docker: <code>docker compose -f deploy/docker-compose.yml ps</code>
The queue is empty	Check that the detection tools are sending to the broker with an ingest key. Demo: rerun <code>start-demo</code> , or on Windows run <code>orchestrator\trigger-alert.ps1</code>
New incidents stopped getting decisions	The analysis queue may hold failed jobs. Ask the administrator to check <code>GET /api/queue</code> on the broker. Detections are stored regardless and can be retried
Every decision says Rule fallback	The model is not answering. Check the LLM dot and System Health; make sure Ollama is running and the model is pulled
An action shows Failed	The response tool refused it. The reason is in the delivery receipt. Fix it in that tool, or edit the plan
An action shows Blocked	The target is not valid for that action, or no connector performs it. Edit the plan and correct the target
Everything shows Dry run	The deployment is still in dry-run mode (<code>RESPONSE_DRY_RUN=true</code>). This is by design until the pilot runbook's Stage 2
A case will not change state	That transition is not allowed. Only the buttons shown are valid next states
Pipeline dots are grey (<code>UNKNOWN</code>)	The health check has not answered yet. Wait 5 seconds or reload. If they stay grey, check that the broker is running

When you report a problem, include the **version** from the footer, the **incident id** (for example `ALT-8AE6D52F8208`), and if possible the **Download JSON** file from the Audit trail.

Change log

Version	Date	What changed	Author
1.0	Summer 2026	First edition, for <code>ao-soc</code> 2.8.8: installation (demo and Docker), sign-in, every menu and page, the Tier-2 workflow (approve, edit, reject, audit, outcome, rollback), Live Alerts playbook, daily routine, glossary and troubleshooting. 31 screenshots from the demo stack	J.Ekrami, Claude (Opus 5)